

Roll No.....Kavyansh

I BE CLASS TEST-II APRIL 2024
Electronics & Telecommunication
1AMRC1 Applied Mathematics-I

Time-70 mins.

Maximum Marks [20]

Note-Attempt all the questions.

Q.1 If $u = \cos^{-1} \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)^{1/2}$, prove that- [4]

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{12} \left(\frac{13}{12} + \frac{\tan^2 u}{12} \right).$$

Q.2 The torsional rigidity of a length of wire is obtained from the formula $N = 8\pi Il / r^2 t^4$. If l is decreased by 2%, r is increased by 2%, t is increased by 1.5%, show that the value of N is diminished by 12% approximately. [4]

Q.3 The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature on the surface of unit sphere $x^2 + y^2 + z^2 = 1$. [4]

Q.4 State green's theorem and evaluate

$$\oint_C [(x^2 + y^2)dx - 2xydy],$$

where C is the rectangle enclosed by $y = 0, y = b, x = 0, x = b$. [4]

Q.5 Evaluate

$$\iint_S (yz\hat{i} + zx\hat{j} + xy\hat{k})ds,$$

where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the first octant. [4]
